2023-2024

S4 MATH CP

PAPER I

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2023-2024 Secondary Four First Term Examination

MATHEMATICS Compulsory Part

PAPER I

Question-Answer Book

04/01/2024

8:15 am – 10:30 am (2¼ hours)

This paper must be answered in English

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.

2. This paper consists of THREE sections, A(1), A(2) and B.

3. Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

4. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string INSIDE this book.

5. Unless otherwise specified, all working must be clearly shown.

6. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.

7. The diagrams in this paper are not necessarily drawn to scale.

Name: _____

Class: _____( )

	Marker Use Only
Question	Mark
1	
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Total			

SECTION A(1) (35 marks)

1. Make y the subject of the formula $x = \frac{3y+4}{5-2y}$. (3 marks)

2. Simplify $\frac{m^{-2}n^5}{(m^3n)^{-4}}$ and express your answer in positive indices. (3 marks)

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3. Factorize

(a) $6x^2 - xy - 2y^2$,

(b) $14x + 7y - 6x^2 + xy + 2y^2$. (3 marks)

4. (a) Round up 4830.297 to the nearest integer.

(b) Round down 4830.297 to 1 significant figure.

(c) Round off 4830.297 to 2 decimal places.

(d) Round off 4830.297 to 3 significant figures. (4 marks)

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5. Find the unknown constants P and Q such that $(x + P)^2 - 3(x - 1) + Q \equiv (x - 1)(x + 2) + 5$.

(5 marks)

6. The number of comics books owned by Anson is 5 times that owned by Joyce. If Anson gives 38 comics books to Joyce, then the number of comics books owned by Joyce is 4 times that owned by Anson. Find the number of comics books owned by Joyce. (4 marks)

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7. Use long division to find the quotient and the remainder when $4x^3 - 11x - 6$ is divided by $x^2 + 3x$.

(4 marks)

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8. If $f(x) = 2x^2 - x + 3$, find the value of $f(2) + f(-3)$. (4 marks)

9. The equation $9x^2 - 42x + c = 0$ has equal roots, where c is a constant.

(a) Find c.

(b) Solve the equation $9x^2 - 42x + c = 25$. (5 marks)

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SECTION A(2) (35 marks)

10. Let $f(x) = x^2 + x - 3$. It is given that $f(-k) = k$, where k is a constant. Find the values of k. (5 marks)

11. Consider the graph of the quadratic function $y = 4(a - x)^2 - 1$, where a is a constant and $a > 0$.

(a) Write down the direction of opening of the graph. (1 mark)

(b) Write down the optimum value of y. (2 marks)

(c) If the y-intercept of the graph is 35, find the value of a. Hence write down the equation of the axis of symmetry of the graph. (4 marks)

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12. A straight line L passes through the point $(4, -2)$ and is perpendicular to the straight line $2x - 5y + 7 = 0$.

(a) Find the equation of L. (4 marks)

(b) Find the x-intercept and the y-intercept of L. (3 marks)

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13. Find the maximum value and the minimum value of $3 \sin^2 x - 1$, where $0° \leq x \leq 360°$. (3 marks)

14. Let $f(x) = x^2 - 3x + 5$. A student claims that $f(3 - k) = f(k)$, where k is a constant. Is the claim correct?

Explain your answer. (4 marks)

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15. The polynomial $p(x)$ is divisible by $x - 4$. When $p(x)$ is divided by $2x^2 - 7$, the quotient and the remainder are $x^2 - 3x - 5$ and $bx - 3b + 1$ respectively, where b is a constant.

(a) Find b. (4 marks)

(b) Prove that $x + 3$ is a factor of $p(x)$. (2 marks)

(c) Someone claims that all the roots of the equation $p(x) = 0$ are real numbers. Is the claim correct? Explain your answer. (3 marks)

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SECTION B (35 marks)

16. It is given that $2\tan(180° + \theta) = 3$, where $0° \le \theta \le 90°$. Find the value of

$$\frac{2\sin(180° - \theta) + \cos(360° + \theta)}{\cos(90° - \theta) - \cos(-\theta)}.$$

(6 marks)

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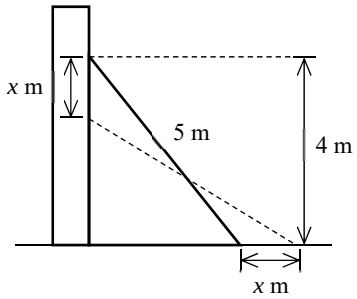
17. Solve the equation $2 \tan x - 3 \sin x = 0$, where $0° \leq x \leq 360°$.

(Give the answers correct to the nearest 0.1° if necessary.) (7 marks)

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18. A ladder with length 5 m leans against a vertical wall. The top of the ladder is 4 m above the horizontal ground.



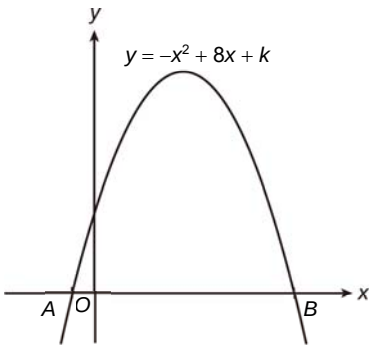
(a) Find the distance between the bottom of the ladder and the wall. (3 marks)

(b) The bottom of the ladder is then pulled away from the wall further for x m. If the top of the ladder slides down by the same distance, find the value of x. (4 marks)

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19. The figure shows the graph of $y = -x^2 + 8x + k$, where k is a constant. One of the x-intercepts is -1.



(a) Find the value of k. (1 mark)

(b) Find the other x-intercept. (2 marks)

(c) What are the coordinates of the vertex of the graph? (2 marks)

(d) A point $P(x, y)$ is moving along the graph of $y = -x^2 + 8x + k$ between A and B. Let $F(x)$ be the function representing the area of ΔABP. Find the maximum area of ΔABP. (2 marks)

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20. The straight line L: $4x - 3y + 12 = 0$ cuts the y-axis and x-axis at the points A and B respectively. C is a point in quadrant IV (i.e., its x-coordinate being positive, while its y-coordinate is negative) such that $AB \perp AC$ and $AC = 10$ units.

(a) Write down the coordinates of A and B. (2 marks)

(b) Find the equation of the straight line passing through A and C. (2 marks)

(c) What is the area of ΔABC? (2 marks)

(d) Suppose that P is a point on AC such that $AP : PC = r : 1$ and the area of ΔBPC is 10 sq. units. Jason claims that he can find r without getting the coordinates of C. Is the claim correct? Explain your answer. (2 marks)

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